

Problem Set 2 - Answer Sheet

Applied Stats II - Nour Kebbi - 23350337

Due: February 18, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in **R**, please include the code you used to get your answers. Please also include the **.R** file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub in **.pdf** form.
- This problem set is due before 23:59 on Wednesday February 18, 2026. No late assignments will be accepted.

We're interested in what types of international environmental agreements or policies people support (Bechtel and Scheve 2013). So, we asked 8,500 individuals whether they support a given policy, and for each participant, we vary the (1) number of countries that participate in the international agreement and (2) sanctions for not following the agreement.

Load in the data labeled **climateSupport.RData** on GitHub, which contains an observational study of 8,500 observations.

- Response variable:
 - **choice**: 1 if the individual agreed with the policy; 0 if the individual did not support the policy
- Explanatory variables:
 - **countries**: Number of participating countries [20 of 192; 80 of 192; 160 of 192]
 - **sanctions**: Sanctions for missing emission reduction targets [None, 5%, 15%, and 20% of the monthly household costs given 2% GDP growth]

Please answer the following questions:

1. Remember, we are interested in predicting the likelihood of an individual supporting a policy based on the number of countries participating and the possible sanctions for non-compliance.

Fit an additive model. Provide the summary output, the global null hypothesis, and p -value. Please describe the results and provide a conclusion.

```
1 load(url("https://github.com/ASDS-TCD/StatsII_2026/blob/main/datasets
  /climateSupport.RData?raw=true"))
2 #ls()
3 #class(climateSupport)
4 df <- as.data.frame(climateSupport)
5 summary(df)
6 #factoring variables
7 df$countries <- as.factor(df$countries)
8 df$sanctions <- as.factor(df$sanctions)
9 #question 1
10 predictive_model <- glm( #this is the predictive model
11   choice ~ countries + sanctions,
12   data = df,
13   family = binomial(link = "logit")
14 )
15 summary(predictive_model)
16 null_model <- glm(choice ~ 1, data = df, family = binomial) #null
  model
17 #now to get the likelihood model we run anova
18 anova(null_model, predictive_model, test = "LRT")
```

The p value is very small, smaller than significance level of 0.05, then we reject the null hypothesis. The Deviance is 215.15, which is the improvement in fit when adding our variable to the null model.

2. If any of the explanatory variables are statistically significant in this model, then:

- (a) For the policy in which nearly all countries participate [160 of 192], how does increasing sanctions from 5% to 15% change the odds that an individual will support the policy? (Interpretation of a coefficient)

```
1 #part a
2 #new data frames
3 data_5_percent_160of192 <- data.frame(countries = factor("160 of 192"
4   , levels = levels(df$countries)),
5   sanctions = factor("5%", levels =
6     levels(df$sanctions)))
7 data_15_percent_160of192 <- data.frame(countries = factor("160 of 192"
8   , levels = levels(df$countries)),
9   sanctions = factor("15%", levels =
10     levels(df$sanctions)))
11 #prediction
```

```

8 log_odds_5_percent <- predict(predictive_model, newdata = data_5_
  percent_160of192, type = "link")
9 log_odds_15_percent <- predict(predictive_model, newdata = data_15_
  percent_160of192, type = "link")
10 change_in_log_odds <- log_odds_15_percent - log_odds_5_percent
11 odds_ratio <- exp(change_in_log_odds)

```

The odds ratio for data section 5 percent to 15 percent sanctions at 160 countries is 0.722. An odds ratio > 1 means the odds increase.

- (b) For the policy in which very few countries participate [20 of 192], how does increasing sanctions from 5% to 15% change the odds that an individual will support the policy? (Interpretation of a coefficient)

```

1 #new data frames
2 data_5_percent_20of192 <- data.frame(countries = factor("20 of 192",
  levels = levels(df$countries)),
3                                     sanctions = factor("5%", levels
  = levels(df$sanctions)))
4 data_15_percent_20of192 <- data.frame(countries = factor("20 of 192",
  levels = levels(df$countries)),
5                                     sanctions = factor("15%",
  levels = levels(df$sanctions)))
6 #prediction
7 log_odds_5_percent_20 <- predict(predictive_model, newdata = data_5_
  percent_20of192, type = "link")
8 log_odds_15_percent_20 <- predict(predictive_model, newdata = data_15_
  percent_20of192, type = "link")
9 change_in_log_odds_20 <- log_odds_15_percent_20 - log_odds_5_percent_
  20
10 odds_ratio_20 <- exp(change_in_log_odds_20)

```

The odds ratio for data section 5 percent to 15 percent sanctions at 20 countries is 0.722. An odds ratio > 1 means the odds increase.

- (c) What is the estimated probability that an individual will support a policy if there are 80 of 192 countries participating with no sanctions?

```

1 #new data frames
2 data_80 <- data.frame(countries = factor("80 of 192", levels = levels
  (df$countries)),
3                       sanctions = factor("None", levels =
  levels(df$sanctions)))
4 #prediction
5 predicted_probability <- predict(predictive_model, newdata = data_80,
  type = "response")

```

The estimate probability for 80 countries with no sanctions is 0.516.

3. Would the answers to 2a and 2b potentially change if we included an interaction term in this model? Why?

- Perform a test to see if including an interaction is appropriate.

```

1 #fitting a model with interaction
2 interaction_model <- glm(
3   choice ~ countries * sanctions, #* for interaction
4   data = df,
5   family = binomial(link = "logit")
6 )
7 #comparing the two models
8 interaction_test <- anova(predictive_model, interaction_model, test =
9   "LRT")
9 print(interaction_test)

```

Pvalue is 0.39 which is higher than 0.05 so we fail to reject null hypothesis.